Predictive Storage Caching and Tiered Data Migration

Integrated with Astra Sim2

DLIO Trace (Parquet)

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chakra_converter

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Chakra ET (Protobuf)

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AstraSim 2 Simulator

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PSC (Tier Decision)

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Cycle-Level Metrics

Converting DLIO Traces → Chakra Execution Traces

Content:

- Built custom converter:
  - Parquet → Protobuf (Chakra ET)

Mapping:

- `read` → `MEM_LOAD_NODE`
- `write` → `MEM_STORE_NODE`
- `size_kb` → `tensor_size (bytes)`
- `phase` → custom metadata field

Injected **ML-predicted phase information into hardware traces**

Execution Graph Representation

- Each I/O operation becomes a node
- Node types used:
  - Memory Load
  - Memory Store

Node contains:

- Operation type (load/store)
- Data size
- Execution order
- Phase (custom attribute)

We convert storage traces into a hardware-executable graph

Bridging to Hardware Simulation

- Wrapper built to:
  - Run AstraSim simulations
  - Parse cycle-level outputs

Hardware modeled:

- 4-tier memory hierarchy:
  - HBM
  - CXL
  - SSD
  - Cold storage

Multi-device simulation:

- Simulated TPU v3-8 using:
  - Trace replication (8 NPUs)

Phase-Conditioned Tier Placement

For each memory request:

1. AstraSim encounters memory node
2. Calls memory system
3. PSC decides: Which tier to place/access data

PSC Input:

- Phase
- Access size
- Context

Output:

- Tier selection (HBM / CXL / SSD / Cold)

End-to-End Evaluation Setup

- Combines:
  - PSC model
  - AstraSim cycle outputs

Baselines added:

- LRU
- LFU

Metrics:

- Total cycles
- Hit rate
- Latency improvements

Performance Gains with PSC

Workload	Metric	Result
ResNet	Speedup vs LRU	30× faster
UNet3D	Cycles Saved	~445M saved
BERT	Hit Rate	84.7% vs 0%